



Daffodil
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ASSIGNMENT

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BST Usages in Banking Management System

Introduction

In today's digital era, banking systems handle a massive amount of customer information and financial transactions every second. Banks must manage customer accounts, transaction records, loan information, and account balances efficiently. To achieve fast data processing and retrieval, various data structures are used in banking software. One of the most important data structures is the **Binary Search Tree (BST)**.

A Binary Search Tree is a hierarchical data structure that stores data in a sorted manner. It allows efficient searching, insertion, and deletion operations. Because banking systems frequently need to search for customer accounts, update account information, and maintain organized records, BST can play a significant role in improving system performance.

This assignment discusses the concept of Binary Search Trees and explains how BST can be used in banking management systems. It also highlights the advantages, limitations, and practical applications of BST in modern banking operations.

What is a Binary Search Tree (BST)?

A Binary Search Tree (BST) is a special type of binary tree where each node contains a value and follows specific rules:

1. All values in the left subtree are smaller than the node value.
2. All values in the right subtree are greater than the node value.
3. Both left and right subtrees must also satisfy BST properties.

A BST consists of three main components:

- Root Node
- Left Child
- Right Child

Example

50

/ \

30 70

/\ /\

20 40 60 80

In this example:

- Values smaller than 50 are placed on the left side.
- Values greater than 50 are placed on the right side.

This structure enables faster searching compared to a simple linear list.

Importance of BST in Banking Systems

Banking systems process thousands or even millions of customer records daily. These records must be accessed quickly and accurately. BST provides an organized way to store data so that information can be searched efficiently.

Some common banking operations include:

- Searching customer accounts
- Opening new accounts
- Updating account information
- Deleting inactive accounts
- Maintaining transaction records

BST helps perform these operations efficiently by reducing the number of comparisons required during searching.

For example, if a bank stores customer account numbers in a BST, the system can locate a specific account much faster than searching sequentially through all records.

Applications of BST in Banking Management System

1. Customer Account Management

One of the primary applications of BST is managing customer accounts.

Each customer account can be stored as a node where:

- Account Number acts as the key.
- Customer details are stored as data.

Example:

```

      5000
    /  \
   /    \
  3000  7000
 /  \  /  \
2000 4000 6000 8000

```

If a customer wants to access account number 6000, the system can quickly find it by following BST rules.

Benefits:

- Fast account search
- Efficient record organization
- Reduced processing time

2. Transaction Record Management

Banks maintain records of deposits, withdrawals, and transfers.

BST can be used to store transaction IDs in sorted order. This enables the system to:

- Find a specific transaction quickly
- Display transactions in sorted order
- Verify transaction history efficiently

For example, when a customer requests a transaction statement, BST helps retrieve records faster than an unsorted structure.

3. Loan Information Management

Banks handle numerous loan applications and records.

A BST can store:

- Loan IDs
- Customer IDs
- Loan amounts
- Approval status

When a bank officer searches for a particular loan application, BST enables rapid access to the required information.

Advantages include:

- Faster loan record retrieval
- Better organization of applications
- Efficient updates of loan status

4. ATM Account Verification

ATM systems need to verify account information instantly.

When a customer inserts a card and enters an account number, the system must quickly determine whether the account exists.

Using BST:

1. The account number is searched.
2. Verification is performed.
3. Account details are displayed.

This reduces waiting time and improves customer experience.

5. Online Banking Services

Modern online banking systems provide services such as:

- Balance inquiry
- Fund transfer
- Account updates
- Transaction history

BST helps organize account information and allows quick retrieval of customer data.

As online banking platforms must respond within seconds, efficient data structures like BST contribute to better performance.

6. Managing Fixed Deposits and Savings Accounts

Banks maintain various account categories.

BST can store:

- Fixed Deposit Accounts
- Savings Accounts
- Current Accounts

The system can quickly search and update records whenever customers modify their banking services.

This improves administrative efficiency and reduces processing delays.

Operations of BST in Banking Systems

Search Operation

Searching is the most frequently used operation in banking applications.

Example:

Searching account number 60:

50

\

70

/

60

Steps:

1. Compare with 50.
2. Move right to 70.

3. Compare with 70.
4. Move left to 60.
5. Record found.

Time complexity:

- Average Case: $O(\log n)$
 - Worst Case: $O(n)$
-

Insertion Operation

When a new customer opens an account, the account information is inserted into the BST.

Example:

Adding account number 65:

```
50
 \
 70
 /
60
 \
 65
```

Benefits:

- Maintains sorted order
 - Supports future searches efficiently
-

Deletion Operation

Sometimes accounts are closed or removed from the database.

BST supports deletion while preserving the tree structure.

Deletion helps:

- Remove inactive records
 - Maintain database cleanliness
 - Improve storage management
-

Advantages of Using BST in Banking Management

1. Fast Searching

BST allows quick retrieval of account information.

2. Efficient Data Organization

Records remain sorted automatically.

3. Easy Data Updates

New records can be inserted without reorganizing the entire database.

4. Better Resource Utilization

BST reduces unnecessary comparisons and processing time.

5. Scalability

It can handle large numbers of customer records effectively.

6. Improved Customer Service

Faster searching means quicker service and better customer satisfaction.

Limitations of BST

Although BST offers many advantages, it also has some limitations.

1. Unbalanced Trees

If data is inserted in sorted order, the BST may become skewed.

Example:

10

\

20

\
30
\
40

In this case, searching becomes slower.

2. Performance Degradation

A skewed BST may perform similarly to a linked list.

3. Memory Overhead

Each node requires additional memory for left and right pointers.

To overcome these issues, balanced BSTs such as AVL Trees and Red-Black Trees are often used in professional banking systems.

Real-World Banking Scenario

Consider a bank with 100,000 customer accounts.

Without BST:

- The system may need to search many records sequentially.

With BST:

- The system narrows down the search path at each step.
- Account information can be found much faster.

For example:

A customer enters account number 7500.

Instead of checking every account, BST follows the appropriate branches until the desired account is found.

This saves time and improves overall banking efficiency.

Future Scope of BST in Banking

As banking systems continue to evolve, efficient data structures remain important.

Future applications may include:

- AI-based banking systems
- Fraud detection mechanisms
- Digital wallet management
- Large-scale customer databases
- Automated loan approval systems

BST and its advanced forms can contribute significantly to these applications by ensuring efficient data storage and retrieval.

Conclusion

Binary Search Tree (BST) is an important data structure that can significantly improve the efficiency of banking management systems. It provides fast searching, insertion, and deletion operations, making it suitable for managing customer accounts, transaction records, loan information, ATM verification, and online banking services.

Although BST has some limitations, its advantages outweigh its drawbacks in many applications. Modern banking systems often use advanced versions of BST to achieve even better performance. Therefore, understanding BST and its practical applications is essential for designing efficient banking software and database systems.

In conclusion, Binary Search Trees play a valuable role in banking management by enabling faster data access, improving system organization, reducing processing time, and enhancing customer service. As banking technology continues to advance, BST-based concepts will remain fundamental in building reliable and efficient banking solutions.